Jonathan A. Klein

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EXPERIENCE

Senior Software Engineer, Cimpress, Remote **Jan 2026 – Present**

- Own core components of in-house manufacturing software, powering **$1.8B+** in annual revenue
- Designed and led development of an in-house AI support pipeline using AWS Bedrock, automating workflows and reducing support requests by **50%**
- Deployed SQS and Lambda via infrastructure-as-code to handle high-throughput traffic, cutting request loss by **99%**

Software Development Engineer, Amazon, New York, NY **Jun 2025 – Jan 2026**

- Owned full-stack development and maintenance of ad management platform powering over **$3B** in annual revenue
- Launched self-serve revision APIs used by **150+** internal advertisers, reducing manual campaign updates by **70%** and cutting internal onboarding time by **40%**
- Built an internal chatbot using AWS Lambda and Bedrock to support team collaboration and employee onboarding
- Participated in a **24/7** on-call rotation, resolving production incidents and ensuring high availability of services

Head Teaching Assistant, Lehigh University, Bethlehem, PA **Jan 2024 – May 2025**

- Created assignments for Data Structures & Algorithms and Operating Systems courses (**350+** students total)
- Led recitation sessions and **8** office hours every week, providing personalized coding instruction and support
- Managed a team of **15** graders and coordinating the workflow for **30+** assignments

Researcher, Scalable Systems & Software Group, Bethlehem, PA **Dec 2022 – Nov 2024**

- Implemented CUDA-based GPU parallelization for zero-knowledge proof generation, achieving over **200%** speedup
- Optimized concurrent data structures in C++ by reducing lock contention and improving cache locality

Software Engineering Intern, FinFolio, Remote **May 2024 – Aug 2024**

- Refactored **50+** SQL queries and reviewed .NET code, improving financial report generation time by over **30%**
- Conducted root cause analysis to diagnose and resolve server-side bugs, enhancing system reliability

Data Science Intern, Mutual of America, New York, NY **Jun 2022 – Jul 2022**

- Implemented natural language processing (NLP) algorithms in Python to find correlations to client conversion rate
- Engineered pipelines to clean and process data, improving accuracy to over **90%** and reducing latency by **30%**

PROJECTS

Sockfish | GitHub

- Engineered a low-latency chess engine using C++, featuring efficient board representation and move generation
- Accelerated graph search (iterative deepening, alpha-beta pruning), speeding up AI move selection by over **500%**
- Created a graphical user interface (GUI) using SFML for an intuitive user experience

Airlytics | AWS Hackathon Winner | Live Demo

- Built a platform displaying global environmental factors on an interactive map using React
- Integrated REST APIs to fetch real-time air quality and pollution data from **250+** cities

UniMart | AWS Hackathon Winner | GitHub

- Launched a campus-exclusive e-commerce platform, promoting sustainability through peer-to-peer item exchange
- Engineered an ID card recognition service with Rekognition, S3, and DynamoDB, improving login time by over **70%**
- Leveraged Firebase for secure user authentication and real-time database management for **750+** users

SKILLS

- **Programming:** C++, TypeScript, Java, C#, SQL, Python, Rust, CUDA
- **Frameworks & Databases:** React.js, Angular, Node.js, .NET, Snowflake, MS SQL Server, MongoDB
- **Infra, DevOps, & Systems:** AWS (ECS, SQS, Lambda, Bedrock, S3, CloudFormation), Linux, Docker, Git, Bash

EDUCATION

Georgia Institute of Technology, Atlanta, GA **Expected Graduation: Aug 2028**
Master of Science in Computer Science, Concentration in Machine Learning **GPA: 4.0 / 4.0**
Lehigh University, Bethlehem, PA **Aug 2021 – May 2025**
Bachelor of Science in Computer Science & Mathematics **GPA: 3.96 / 4.0**

- **Relevant Courses:** Data Structures & Algorithms, Artificial Intelligence, Operating Systems, Parallel Computing, Databases, Advanced Algorithms, Programming Languages, Network Security, Graph Theory, Systems Software
- **Honors:** Summa Cum Laude, Upsilon Pi Epsilon, Tau Mu Lambda, Wilbur Mathematics Prize, Dean's List
- **Extracurriculars:** ACM (President), Mathematics & Statistics Society (Vice President), Chess Club